ASTEROSEISMIC ANALYSIS OF 16 CYGNI A USING KEPLER PHOTOMETRIC LIGHT CURVES

A Computational Study of Stellar Oscillations and Their Relationship to Fundamental Stellar Properties

Project Domain: Astronomy, Astrophysics and Scientific Computing
Target Star: 16 Cygni A
Kepler Input Catalog ID: KIC 12069424
Primary Data Source: NASA Kepler Mission
Programming Language: Python

ABSTRACT

Asteroseismology is the study of stellar oscillations to investigate the internal structure and physical properties of stars. These oscillations can produce small variations in observed stellar brightness, which can be analysed using time-series processing and frequency-domain techniques.

This project investigates the solar-like oscillations of 16 Cygni A (KIC 12069424) using photometric observations from NASA's Kepler mission. The study will involve obtaining public light-curve data, examining data quality, preprocessing the observations and analysing their frequency content using a power spectrum. Candidate oscillation frequencies and global asteroseismic parameters will be investigated where the data support reliable measurements.

The extracted parameters will be compared with published measurements for 16 Cygni A to evaluate the reliability of the computational workflow. Uncertainties arising from instrumental noise, data gaps, finite observation duration and processing choices will also be considered.

The expected outcome is a reproducible Python based workflow demonstrating how photometric observations can be transformed into meaningful asteroseismic information.

1. INTRODUCTION

Stars are dynamic physical systems whose internal structures influence their observable properties. Although a star may appear nearly constant in brightness, its surface can undergo small variations caused by internal oscillations, convection and other physical processes.

Asteroseismology uses these oscillations to investigate stellar interiors. Similar to how the sound produced by a musical instrument depends on its physical structure, the oscillation frequencies of a star depend on properties such as its density, internal sound-speed profile and evolutionary state.

Photometric observations provide a way to investigate these oscillations without directly observing the motion of individual regions inside the star. By recording brightness over time and analysing its frequency content, researchers can identify characteristic oscillation patterns.

NASA's Kepler mission obtained high precision photometric observations of numerous stars, making its public data valuable for asteroseismic research.

This project focuses on 16 Cygni A, a well-studied solar-like oscillator with published asteroseismic measurements. These measurements provide reference values against which the results of the computational analysis can be evaluated.

2. RESEARCH QUESTION AND OBJECTIVES

2.1 Research Question

Can small periodic variations in the observed brightness of 16 Cygni A be analysed to identify stellar oscillation frequencies, and how closely do the extracted frequency patterns agree with published asteroseismic measurements?

2.2 Objectives

The objectives of this project are:

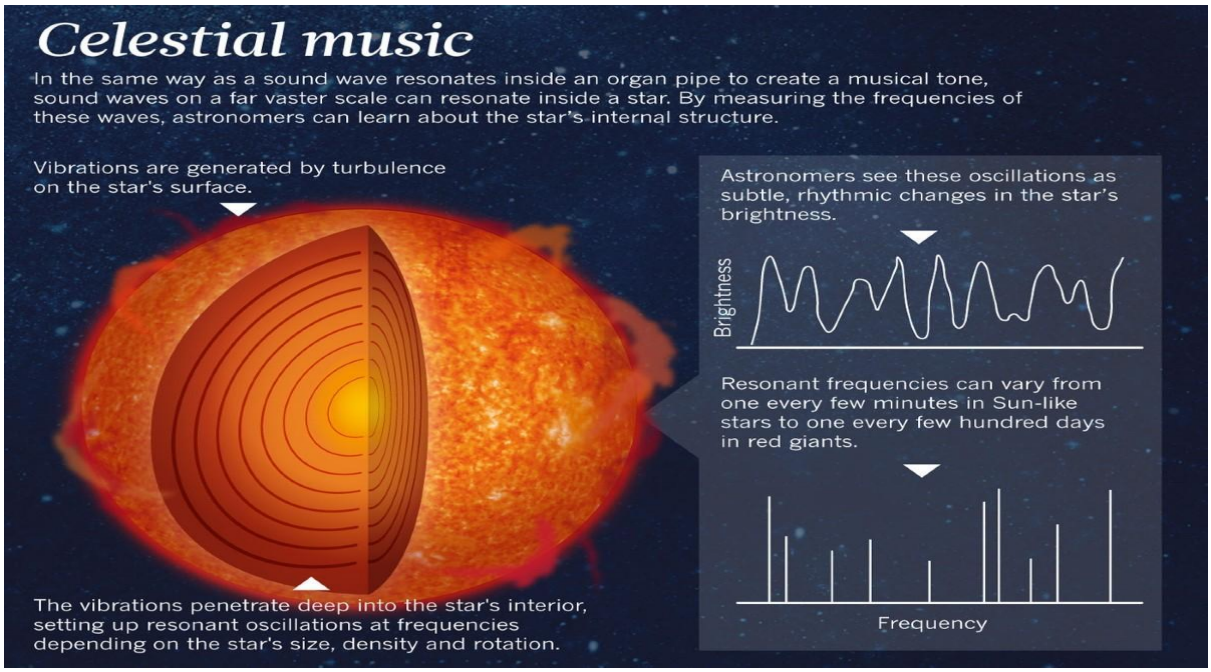
1. To obtain publicly available Kepler photometric observations of 16 Cygni A.
2. To inspect and preprocess the light-curve data.
3. To construct a power spectrum for investigating the frequency content of the observations.
4. To identify candidate oscillation peaks and estimate their frequencies.
5. To investigate the frequency of maximum oscillation power and the large frequency separation, where reliable estimates are possible.
6. To compare the measured parameters with published reference values.
7. To assess uncertainty, noise, data-processing effects and limitations.
8. To develop a reproducible Python workflow for computational asteroseismology.

2.3 Working Hypothesis

If the available Kepler observations have sufficient quality and duration, their power spectrum should contain frequency-dependent features consistent with the known solar-like oscillations of 16 Cygni A.

The analysis will test this hypothesis rather than assume that the expected features must necessarily be recovered.

3. SCIENTIFIC BACKGROUND



3.1 Stellar Oscillations

Stellar oscillations are motions of stellar material around an equilibrium configuration. They arise through physical processes involving pressure, gravity and buoyancy.

Two important categories of oscillation modes are pressure modes and gravity modes.

- **Pressure modes (p-modes)** are primarily restored by pressure forces. They are commonly investigated in solar-like oscillators and provide information about the structure of stellar envelopes.
- **Gravity modes (g-modes)** are primarily restored by buoyancy. They can provide information about deep stellar interiors, although their detectability depends on the star and the observational method.

Solar-like oscillations are excited and damped by turbulent convection in the outer layers of suitable stars. Their observable signatures can appear as small brightness variations.

3.2 Photometric Light Curves

A photometric light curve is a time series containing measurements of a star's brightness.

It can be represented as

$$F(t) = \text{measured stellar flux at time } t$$

where (F(t)) is the measured flux and (t) is the observation time.

A light curve allows changes in brightness to be examined in the time domain. However, small oscillations may not be visually obvious because the observed signal can also contain noise, instrumental trends and other variability.

Frequency-domain analysis is therefore used to investigate periodic components that may be difficult to identify directly from the original light curve.

3.3 Fourier Analysis and the Power Spectrum

Fourier analysis represents a time-dependent signal in terms of its frequency components.

A simplified continuous Fourier transform is

$$X(f) = \int x\,(t) e^{-2\pi i f t} dt$$

where (x(t)) represents the brightness variation and (f) is frequency.

The power spectrum describes how the signal's power is distributed across frequency. In asteroseismology, oscillation modes may appear as peaks or groups of peaks above the background noise.

The interpretation of these peaks requires care because instrumental effects, sampling and other sources of variability can also produce spectral features.

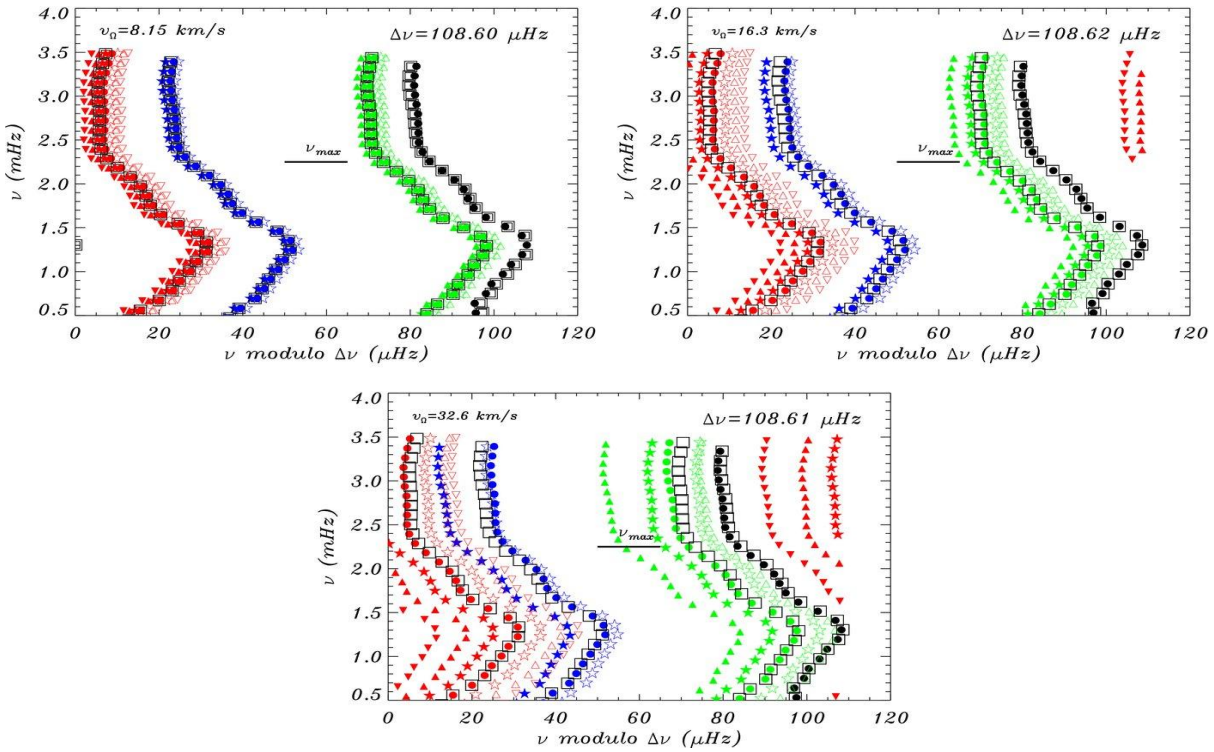
3.4 Important Asteroseismic Parameters

Frequency of maximum oscillation power($\nu_{\max}$): This is the approximate frequency around which the oscillation-power envelope is strongest.

Large frequency separation($\Delta\nu$): This is the approximate frequency spacing between consecutive radial oscillation modes of the same angular degree.

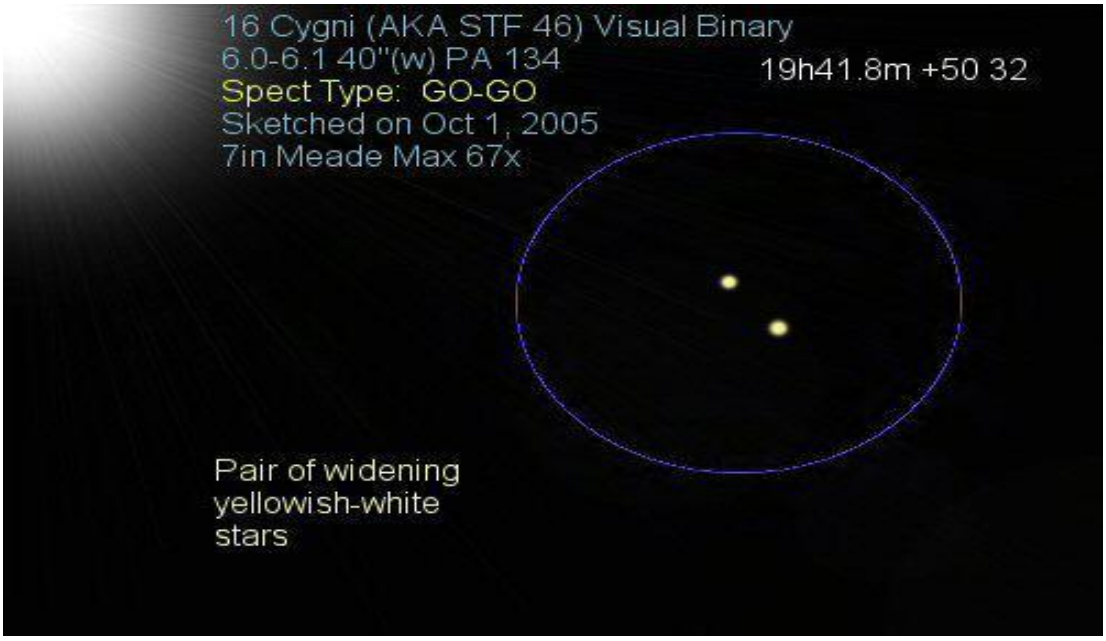
$$\Delta\nu \approx \nu_{n+1,\ell} - \nu_{n,\ell}$$

Here, (n) is the radial order and ℓ is the angular degree of the oscillation mode.



These parameters can be related to stellar properties through asteroseismic models and scaling relations. Their use requires reliable measurements and an understanding of the assumptions and uncertainties involved.

4. TARGET SELECTION: 16 CYGNI A



16 Cygni A is a well-studied solar-like oscillator in the 16 Cygni binary system. Its observations from the Kepler mission and its published asteroseismic measurements make it a useful target for a computational investigation.

A published study combining asteroseismic information with interferometric measurements reported the following reference values for 16 Cygni A:

Parameter	**Published reference value**
Large frequency separation ($\Delta\nu$)	(103.5 \pm 0.1\ \mu\text{Hz})
Frequency of maximum power ($\nu_{\max}$)	(2201 \pm 20\ \mu\text{Hz})
Stellar radius	(1.218 \pm 0.012,R_\odot)

The availability of published reference measurements is particularly useful because the reliability of the computational workflow can be assessed through comparison with independent research.

Reference: White et al. (2013), *Interferometric radii of bright Kepler stars with the CHARA Array: θ Cygni and 16 Cygni A and B*.

5. PROBLEM STATEMENT AND RESEARCH GAP

5.1 Problem Statement

Stellar oscillations produce extremely small brightness variations that may be difficult to distinguish from instrumental noise, long term trends, data gaps and other sources of stellar variability.

A raw light curve does not directly reveal the complete frequency structure of a star. Preprocessing and frequency-domain analysis are required to identify candidate oscillation signatures and investigate their significance.

The central problem addressed by this project is how to transform public photometric observations into interpretable asteroseismic measurements using a transparent and reproducible computational workflow.

5.2 Research Gap

16 Cygni A has already been investigated extensively through asteroseismology and independent observational techniques. Therefore, this project does not claim to discover previously unknown oscillations in the star.

Instead, it addresses a computational and educational objective: demonstrating the complete process of obtaining public observations, processing a light curve, analysing its frequency content and evaluating the results against published measurements. .

The project will examine how data preprocessing influences the visibility of oscillation features, whether the available observations support reliable parameter estimation, and how uncertainty and processing choices affect the final interpretation.

The emphasis is on reproducibility, validation and critical evaluation rather than claiming novelty in the underlying astrophysical discovery.

6. PROPOSED METHODOLOGY

This will follow a sequence of data acquisition, preprocessing, frequency analysis and validation.

- **Step 1: Data Acquisition**
  Search the Mikulski Archive for Space Telescopes (MAST) for available Kepler photometric light-curve products associated with KIC 12069424. Short-cadence observations will be prioritised if suitable products are available.

- **Step 2: Data Inspection**
  Inspect the observation times, flux measurements, cadence, data gaps, quality flags and total observation duration. These factors determine which analyses are appropriate.

- **Step 3: Preprocessing**
  Remove unreliable observations where justified, normalise the flux and examine long-term trends. Any detrending method must be selected carefully to avoid removing genuine stellar variability.
- **Step 4: Frequency Analysis**
  Construct a power spectrum and investigate the distribution of power across frequency. Candidate peaks will be examined in relation to the expected oscillation pattern and the background noise.

- **Step 5: Parameter Estimation**
  Estimate (\nu_{\max}) and investigate $\Delta\nu$ if the observed spectrum supports reliable measurements. More detailed mode identification may be required to obtain a robust large frequency separation.

- **Step 6: Validation**
  Compare the derived values with published measurements for 16 Cygni A. Record the methods, uncertainties and processing assumptions used for each result.

- **Step 7: Interpretation**
  Discuss the evidence for stellar oscillations, the agreement with published results and the limitations imposed by the data and analysis methods.

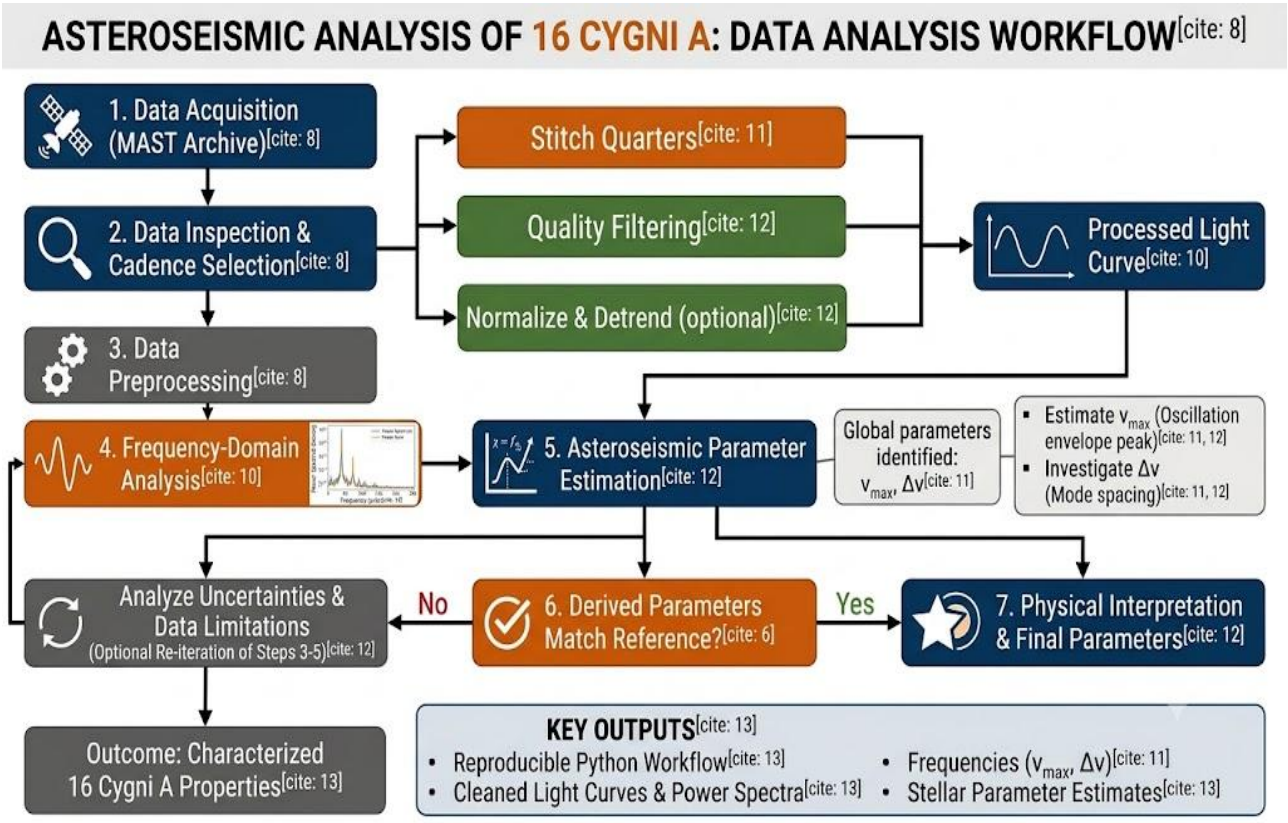
7. DATA SOURCE AND COMPUTATIONAL TOOLS

The primary data source is NASA's Kepler mission, which collected photometric observations of stars over extended periods.

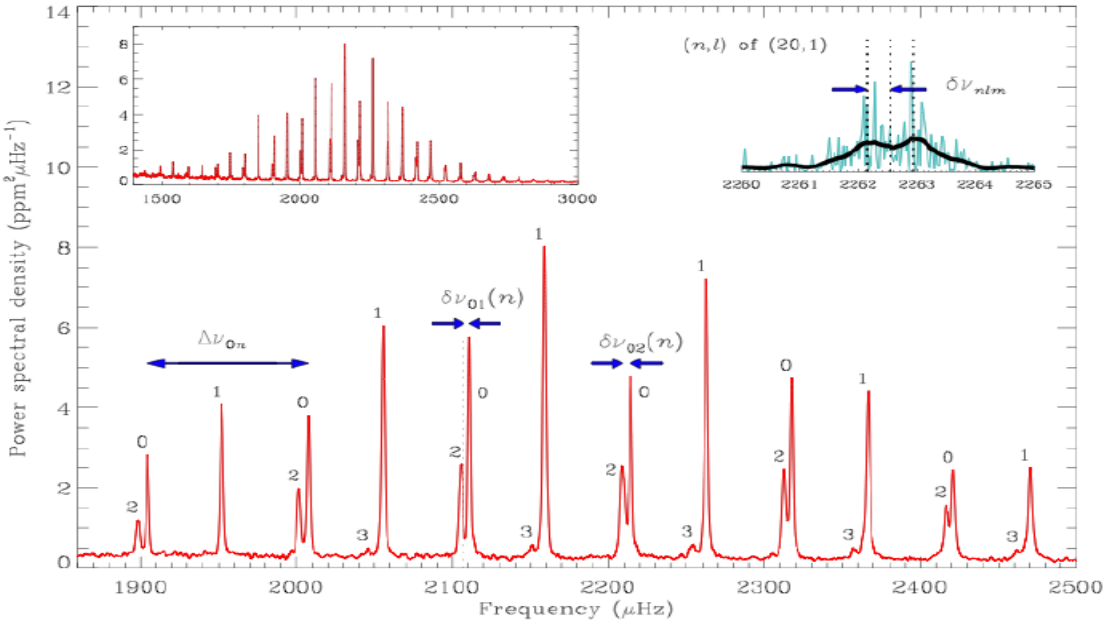
The project will use Python and the following libraries:

- **Lightkurve:** Searching for, downloading and analysing Kepler light curves.
- **NumPy:** Numerical calculations and array operations.
- **SciPy:** Signal-processing and peak-analysis tools.
- **Matplotlib:** Plotting light curves, power spectra and analysis results.
- **Jupyter Notebook:** Running and documenting the analysis interactively.

The availability of suitable data products will be checked before the final analysis. If short-cadence products are not available, the archive will be inspected for other suitable observations before deciding how to proceed.



9. LIGHT-CURVE PREPROCESSING



The initial observations must be examined before frequency analysis is performed.

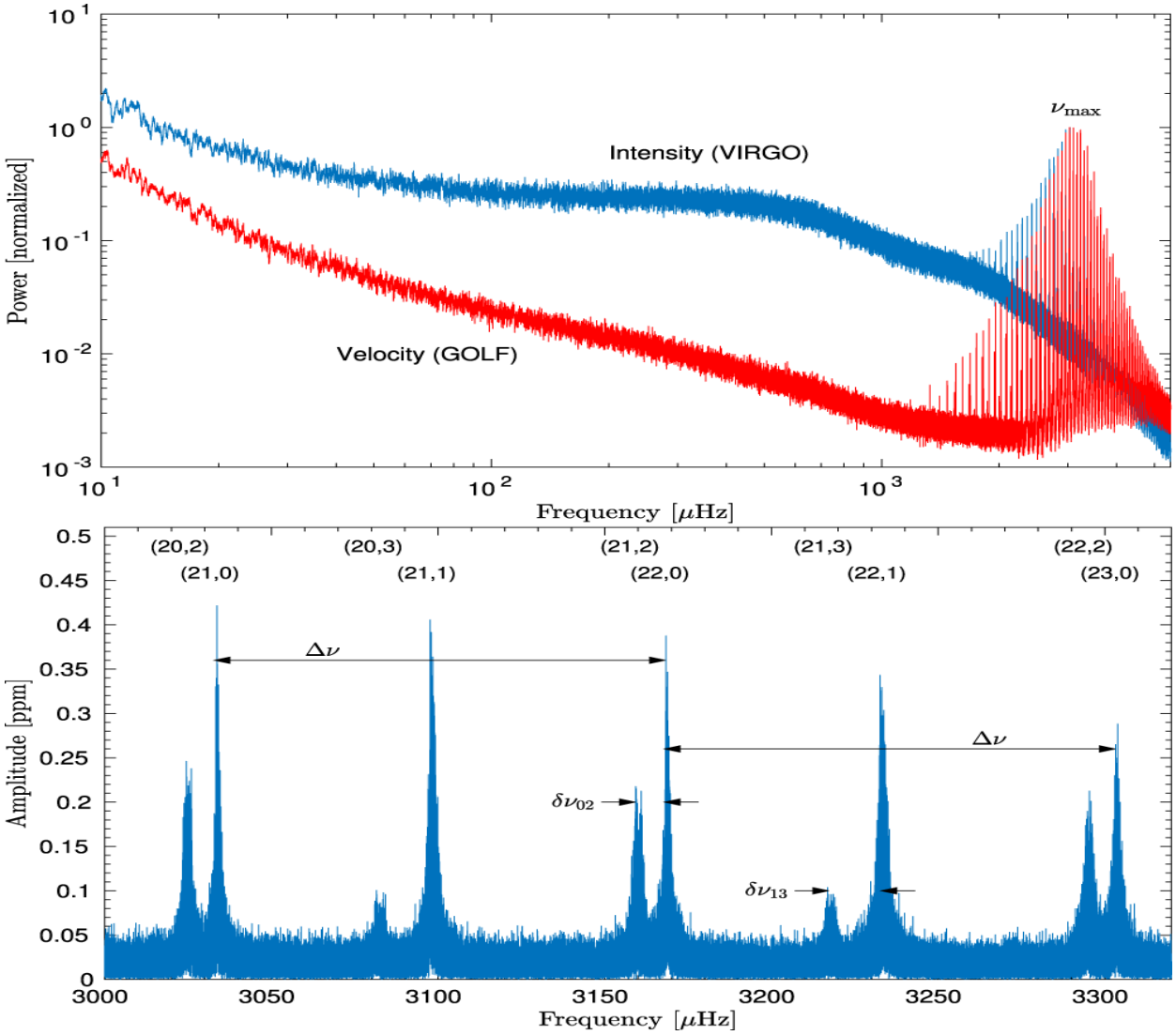
The preprocessing procedure will involve:

1. Inspecting the original light curve.
2. Removing unreliable measurements using appropriate quality information.
3. Removing missing or invalid values.
4. Normalising the flux to make relative brightness variations easier to interpret.
5. Examining long-term trends and removing them only when justified.
6. Identifying data gaps and discontinuities.
7. Comparing the original and processed light curves.

Overly aggressive smoothing or detrending can remove genuine stellar variability. The effects of preprocessing must therefore be considered when interpreting the final power spectrum.

The expected output is a cleaned light curve accompanied by a comparison plot showing the original and processed observations.

10. FREQUENCY-DOMAIN ANALYSIS



Once the light curve has been prepared, its frequency content will be investigated using a power spectrum.

The analysis will focus on the overall frequency distribution, the region containing the strongest oscillation power, candidate individual peaks and the background noise.

The frequency of maximum power $\nu_{\max}$ can be estimated by examining the oscillation-power envelope. The large frequency separation $\Delta\nu$ may be estimated by investigating the spacing between appropriate oscillation modes.

A simple search for the tallest peaks is not sufficient to establish a reliable asteroseismic measurement. The interpretation must consider the expected mode pattern, background noise, frequency resolution and potential instrumental features.

10. PYTHON IMPLEMENTATION

10.1 Searching for Observations

The following code searches for Kepler short-cadence light-curve products associated with 16 Cygni A. The returned products should be inspected to confirm their cadence, observing quarters and availability.

```
import lightkurve as lk
import matplotlib.pyplot as plt

target = "KIC 12069424"

search = lk.search_lightcurve(
    target,
    mission="Kepler",
    exptime="short"
)

print(search)
print(search.table)
```

10.2 Downloading and Inspecting the Data

```
if len(search) == 0:
    raise RuntimeError(
        "No short-cadence products found. "
        "Inspect the archive for other available products."
    )

light_curves = search.download_all()

if light_curves is None or len(light_curves) == 0:
    raise RuntimeError("No light curves were downloaded.")

for lc in light_curves:
    lc.plot()

plt.show()
```

This step checks data access and provides an initial visual assessment of the observations.

10.3 Initial Preprocessing and Power Spectrum

```
import matplotlib.pyplot as plt

# Combine available observations into one time series
lc = light_curves.stitch()

# Remove poor-quality points and normalise flux
lc = lc.remove_nans()
lc = lc.remove_outliers()
lc = lc.normalize()

# Plot the prepared light curve
lc.plot()
plt.show()

# Inspect the power spectrum
pg = lc.to_periodogram(
    method="lombscargle",
    normalization="psd"
)

pg.plot()
plt.show()
```

This code is a starting point and may require adjustments after the actual data products are inspected. Quality flags, gaps, discontinuities and long-term trends should be evaluated carefully before interpreting the spectrum.

The final analysis should document the chosen preprocessing method and confirm the units and frequency range used.

11. VALIDATION AND UNCERTAINTY ANALYSIS

The derived parameters will be compared with published reference values for 16 Cygni A.

The principal comparisons are:

- Measured $\nu_{\max}$ against the published reference value of $\nu_{\max} = 2201 \pm 20$ μHz
- Measured $\Delta\nu$ against the published reference value of ($\Delta\nu = 103.5 \pm 0.1$ μHz
- Any additional stellar parameters only where their estimation is supported by the data and the adopted method.

The uncertainty analysis will consider finite observation duration, frequency resolution, instrumental noise, data gaps, imperfect detrending, peak identification and the influence of analysis settings.

The project will also examine whether reasonable changes in preprocessing or frequency-analysis settings significantly affect the inferred parameters.

A difference between the measured and published values will be investigated in the context of uncertainties and methodology rather than automatically interpreted as a new astrophysical result.

12. LIMITATIONS

Several limitations may affect the reliability of the analysis.

- First, the available data products may differ in cadence, duration and quality. These differences influence the range of frequencies that can be investigated reliably.
- Second, instrumental noise and data gaps can complicate the identification of oscillation peaks.
- Third, the choice of preprocessing method can affect the visibility of low-amplitude signals.
- Fourth, estimating the large frequency separation may require detailed mode identification rather than simply selecting the tallest peaks.
- Finally, converting asteroseismic measurements into estimates of stellar mass, radius or age requires appropriate models or scaling relations and their associated assumptions.

  For these reasons, the project will report only parameters that can be supported by the available observations and will clearly distinguish measured values from published benchmarks.

13. EXPECTED OUTCOME AND SIGNIFICANCE

- The project is expected to demonstrate how public photometric observations can be used to investigate stellar oscillations through computational time-series analysis.
- The main outcomes will be a documented Python workflow, processed light-curve visualisations, a power spectrum, candidate frequency information, uncertainty analysis and comparison with published measurements.
- The project will also demonstrate the importance of preprocessing, reproducibility and independent validation in computational astrophysics.
- Its principal value lies in connecting observational data with physical interpretation while clearly documenting the limitations of the analysis.

14. CONCLUSION

This project investigates the oscillation signatures of 16 Cygni A using photometric observations from NASA's Kepler mission.

By combining light-curve preprocessing, power-spectrum analysis and comparison with published asteroseismic measurements, the study will demonstrate how small brightness variations can provide information about stellar oscillations.

The analysis will focus on identifying candidate oscillation frequencies and investigating global parameters such as $\nu_{\max}$ and $\Delta\nu$ where the data support reliable estimates.

The final conclusions will be based on the actual observations and will account for uncertainties, instrumental effects and the limitations of the adopted methods.

The project will provide a reproducible computational study of a well-characterised solar-like oscillator and demonstrate the role of data analysis in modern astrophysical research.

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By

Name : Anutharini [25BEE0062]